

Can someone intuitively explain what the convolution integral is?

Asked 11 years, 10 months ago Modified 10 months ago Viewed 27k times



23



I'm having a hard time understanding how the convolution integral works (for Laplace transforms of two functions multiplied together) and was hoping someone could clear the topic up or link to sources that easily explain it. Note: I don't understand math notation very well, so a watered-down explanation would do just fine.

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) \, d\tau$$

This is what my textbook has written. What do those lowercase t-like symbols represent (I haven't seen them before).

[ordinary-differential-equations](#) [intuition](#) [convolution](#)

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edited Dec 11, 2012 at 2:03



dtldarek

37.6k

9

56

128

asked Dec 11, 2012 at 1:42



Bailor Tow

575

4

11

22

3 That's a greek letter called tau. It's just a variable here. – [Pedro](#) Dec 11, 2012 at 1:53

2 Lowercase t-like symbol is a greek letter "tau". Here it represents an integration (dummy) variable, which "runs" from lower integration limit, "0", to upper integration limit, "t". So, the convolution is a function, which value for any value of argument (independent variable) "t" is expressed as an integral over dummy variable "tau". – [mbaitoff](#) Dec 11, 2012 at 1:55

1 ... and it's common to choose related names for related variables. Some other examples of letters that are often paired are (z, ζ) , (x, ξ) , (r, ρ) , (s, σ) , (w, ω) , (a, α) , (b, β) , (d, δ) , (e, ϵ) . Also, the letters $\iota, \kappa, \lambda, \mu, \nu$ are often used in places where one might also tend to use i, k, l, m, n respectively. All of the "strange" letters I've used here are from the Greek alphabet. – [user14972](#) Dec 11, 2012 at 1:57

3 Maybe the animated gifs at [Wikipedia](#) will give you some ideas. – [dtldarek](#) Dec 11, 2012 at 2:05

2 I'm late to the party, I know, but I recently stumbled across [Terry Tao's explanation on mathoverflow](#). – [user170231](#) Nov 10, 2015 at 21:46

6 Answers

Sorted by: Highest score (default)



19

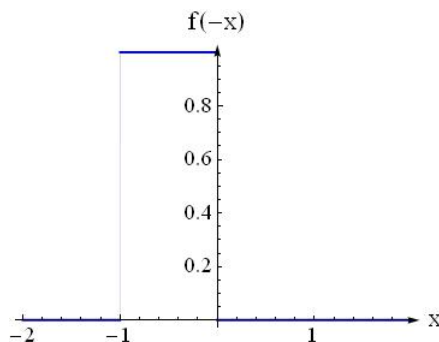


Intuitively speaking when you are given two signals/or functions f and g . You time reverse *one of the signals*, it doesn't matter which one, and shift it by a value of t then you simply multiply and then sum the area under the *intersection*.

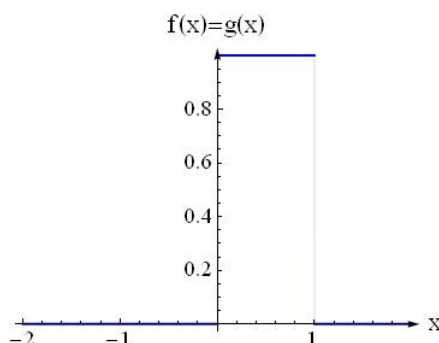
If you consider a function say function of x , then time reversal means inserting $-x$ wherever you see x in this function.

Example:

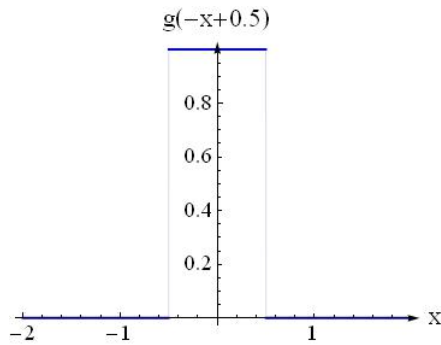
Question1: Assume you have a function $f(x)$ that is 1 if $x \in [0, 1]$, and 0 elsewhere then how should you plot $f(-x)$?



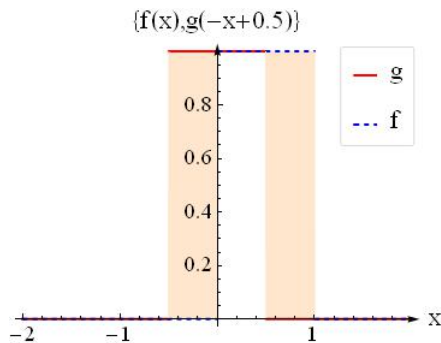
Question2: Assume you have $g(x) = f(x)$ is there any intersecting area between $f(x)$ and $g(x)$?



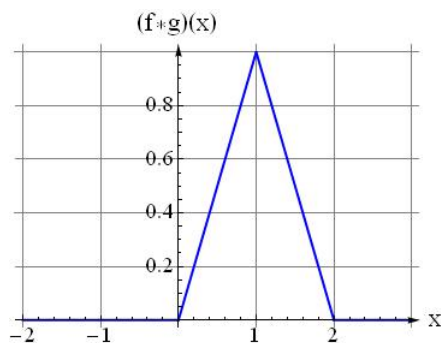
Question3: Now shift $g(-x)$ by 0.5, that is to find $g(-x + 0.5)$. How does it look like when you plot it?



Question4: Where does the intersecting region lie in this case $x \in ?$ what is the are of the intersecting region? Answer: **below white area= 0.5 at $x = 0.5$ shift.**



Question5: If you select the shifting parameter not 0.5 but all reals in $[0, 2]$ what function should you get at the output? **check $x = 0.5$ and see $f * g(x)$ is 0.5 as found at step 4**



EDIT: You define convolution integral in $[0, t]$ for bounded signals. The integral limits depend on where your signal is non-zero.

If you have two signals as you suggested $f(t) = e^{at}$ and $g(t) = e^{bt}$ then the first question: what is the relation between a and b ? are they positive? where is the function defined? For example when a and b are some positive terms then we have the following integral

$$h(t) = \int f(\tau)g(t - \tau)d\tau = \int e^{a\tau}e^{b(t-\tau)}d\tau = e^{bt} \int e^{(a-b)\tau}d\tau = \left[\frac{e^{(a-b)\tau}}{a-b} \right]_{\tau \in \Omega}$$

clearly $\Omega = \mathbb{R}$ is not possible because the integral does not converge.

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edited May 11, 2018 at 13:25

answered Dec 11, 2012 at 2:33

Seyhmus Güngören
7,902 3 27 49

1 Can you use a more concrete example? For example, how can we find the convolution of $f(t) = \sin 2t$ and $g(t) = e^t$ or $f(t) = e^{at}$ and $g(t) = e^{bt}$ with $a \neq b$ in the second example? Or perhaps using the same theorem to find an inverse Laplace of $F(s) = \frac{4}{s(s^2+4)}$? - [Bailor Tow](#) Dec 11, 2012 at 2:36

please see the edit - [Seyhmus Güngören](#) Dec 11, 2012 at 3:10



I believe the convolution functions makes the most sense when you see it applied in probability theory.

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Let X and Y be two random variables and $f(X)$ and $g(Y)$ be the probability distributions of the random variables.



Then the distribution of the sum of two random variables:

$$(f * g)(t) = \int_0^t f(-\tau)g(\tau - t)d\tau$$



Why is this? Let us visualize the simple case of rolling dices, and X be the outcome of the first roll and Y be the outcome of the second roll. What is the distribution of the sum?

Since our distributions are discrete,

$$(f * g)(t) = \sum_{i=0}^t f(i)g(t-i) \quad t \in [2, 12]$$

This basically translates to, sum up all probabilities such that it has this probability.

i.e.

$$(f * g)(4) = \sum_{i=0}^4 f(i)g(4-i) = f(1)g(3) + f(2)g(2) + f(3)g(1) = 1/12$$

Which is the answer we expect. We can also look at the question from a more physics point of view where it is a time reversed signals but I find this much more intuitive.

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answered Dec 11, 2012 at 2:51

 **Andrew**
293 1 8

2  Shouldn't that be $g(t-i)$? And why isn't the convolution integrand $f(\tau)g(t-\tau)$? – Mateen Ulhaq Aug 2, 2015 at 7:15 

1  This result is proved in a "The Fourier Transform and its applications" lecture on YouTube: [youtube.com/watch?v=LA4Uv6PMRTM](https://www.youtube.com/watch?v=LA4Uv6PMRTM) – Pramod Dec 6, 2015 at 23:01 



Consider the sequences $x_0, x_1, \dots$ and $y_0, y_1, \dots$

8

Now,

$$\left(\sum_{j=0}^n x_j \right) \left(\sum_{j=0}^n y_j \right) = \sum_{z=0}^n z_j,$$

where

$$z_j = \sum_{k=0}^j x_k y_{j-k}.$$

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edited Feb 20, 2013 at 21:17

answered Dec 11, 2012 at 2:03

 **André Caldas**
6,151 24 35

 The j subindex on the RHS seems not be bound by anything. – siamii Sep 26, 2016 at 16:10 

 @siamii: The j is fixed on the LHS. For each j you have one z_j . The RHS is how to calculate z_j . – André Caldas Oct 2, 2016 at 3:51 



1

If you want a broad overview, the convolution "blends" two functions together & is the expression of the amount of overlap of one function as it is shifted over another. The convolution takes two functions (& one of them may be a kernel). Writing one of them as a translation, multiply them together & they give you a new function that takes the best properties of both functions. If you take a kernel (as I mentioned above) the new function may have properties from that kernel. A good example would be from something I am interested in: Littlewood-Paley theory. When embarking on various LP constructions, we end up with LP operators such as

$$\Delta_j(f) = \Delta_j^\Psi = \Psi_{2^{-j}} * f.$$

These have been defined by constructing a partition of unity (where we are constructing results locally & extending them globally) where Ψ is a radial Schwartz function on $\mathbb{R}^n$ with certain support properties.

Looking at the convolution, $\Delta_j(f)$ has properties that f had as well as support properties passed onto it from Ψ .

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answered Apr 25, 2015 at 14:31

user230715



1

Intuition for Convolution

A convolution is the amount of an overlap area of one function f as it is shifted over another function g at a given time offset.

Example using discrete valued functions

Let's say we are transforming a certain function $f(t)$ by passing it through a filter $g(t)$ to get the output $h(t)$:

$$f(t) \rightarrow [g(t)] \rightarrow h(t)$$

Say $f(t)$ has the following values for $t = [1, 2, 3, 4, 5] \rightarrow f(t) = [1, 2, 3, 4, 5]$, and

say $g(t)$ has the following values for $t = [1, 2, 3] \rightarrow g(t) = [3, 2, 1]$

Now the question is what is the value of $h(t)$ at a specified time t .

To find the value of $h(t)$ at any time t , let's start with following -

Idea:

Imagine flipping the $f(t)$ with respect to time values, because that is what it looks like when it enters the filter $[g(t)]$

Now let's start here,

->| start of time (t)
[1, 2, 3] -> time

$f(t) = [5, 4, 3, 2, 1]$
 $g(t) = [3, 2, 1]$

To calculate the values of $h(t)$, let's line up the values of $f(t)$ and pass them through the values of $g(t)$

At $t = 1$

$g(t)$	3 2 1	
$f(t)$	5 4 3 2 1	
Total value	3	$\leftarrow (3 \times 1) = 3$

At $t = 2$

$g(t)$	3 2 1	
$f(t)$	5 4 3 2 1	
Total value	6 2	$\leftarrow (3 \times 2) + (2 \times 1) = 8$

At $t = 3$

$g(t)$	3 2 1	
$f(t)$	5 4 3 2 1	
Total value	9 4 1	$\leftarrow (3 \times 3) + (2 \times 2) + (1 \times 1) = 14$

At $t = 4$

$g(t)$	3 2 1	
$f(t)$	5 4 3 2 1	
Total value	12 6 2	$\leftarrow (3 \times 4) + (2 \times 3) + (1 \times 2) = 20$

At $t = 5$

$g(t)$	3 2 1	
$f(t)$	5 4 3 2 1	
Total value	15 8 3	$\leftarrow (3 \times 5) + (2 \times 4) + (1 \times 3) = 26$

At $t = 6$

$g(t)$	3 2 1	
$f(t)$	5 4 3 2 1	
Total value	10 4	$\leftarrow (2 \times 5) + (1 \times 4) = 14$

At $t = 7$

$g(t)$	3 2 1	
$f(t)$	5 4 3 2 1	
Total value	5	$\leftarrow (1 \times 5) = 5$

Just by sliding to any $t = n$, we can find the value of $h(t)$ by calculating the value of overlapped "area" at $t = n$.

Now the value $h(t) = \text{conv}(g(t), f(t))$ at any time t is as follows

$g(t)$	*	$f(t)$	=	$h(t)$
[3 2 1]		[1 2 3 4 5]		[3 8 14 20 26 14 5]
Time	- - - - -	- - - - -	- - - - -	1 2 3 4 5 6 7

Note:

So far we have been doing a simple summation of terms as our functions are discrete. *If we are dealing with **continuous functions**, the integral as a limit of a summation would come into play.*

If you have **Matlab**, please see <https://www.mathworks.com/matlabcentral/fileexchange/4616-animated-convolution>

Credit: The above example is based on the [Intuitive Guide to Convolution @ Better Explained](#)

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edited Nov 30, 2023 at 19:46

answered Sep 30, 2021 at 21:18

 Kokeb
131 2



0



This is quite late to the party but here is another (in my opinion very useful) way to look at the convolution integral: changing frames of reference in Quantum Mechanics.

Suppose I have a quantum particle P_1 whose location relative to a fixed observer O is given by a probability distribution $p_1(x, t)$ (here x is a real number, basically a coordinate on a line) such that given a set of locations D on the line at a particular time t_s the probability of finding the particle in the location D is given by

$$\int_D p_1(x, t_s) dx$$

To be concrete in our 1 dimensional example we might ask "what is the probability of finding the particle in the interval from $x = 1$ to $x = 5$?" and we would find this by evaluating

$$\int_1^5 p_1(x, t_s) dx$$

(for people with a physics background $p_1(x, t) = |\Psi_1(x, t)|^2$).

Now suppose I have another particle P_2 whose location relative to the fixed observer O is given by a similar probability distribution $p_2(x, t)$.

It is natural then to ask, what is the relative location of p_2 versus p_1 ? For example: "what is the probability that p_2 is 3 or more units to the left of p_1 at time t_s ?" In classical mechanics this is easy to do, we just "subtract the frame of reference" but in quantum mechanics this is much trickier (I can elaborate on this if requested in the comments), our desired "frame of reference change" should work as below:

We are looking for a probability distribution function $R(x, t)$ such that if we have a relative space D then

$$\int_D R(x, t_s) dx$$

Gives the probability of P_2 and P_1 in that relative configuration at time t_s . To be concrete we will use the example from earlier:

If we ask "What is the probability that P_2 is 3 or more units to the left of P_1 at time t_s ?" then the answer to that must be

$$\int_{-\infty}^{-3} R(x, t_s) dx$$

Now it turns out after all this set up that

$$R(x, t) = \int_{-\infty}^{\infty} p_2(x - y, t) p_1(y, t) dy$$

Which is just a convolution. So here's yet another application/way to think about the convolution.

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answered Oct 20, 2021 at 18:23

 Sidharth Ghoshal
17.9k 9 44 104